

Xóm Retail

Global Retail Analytics

This project analyzes a multi-country retail dataset and turns the results into an interactive Power BI dashboard for merchandising, store operations, CRM, finance and executive stakeholders.

Dataset	4 tables · 80,734 rows · 33 columns
Dashboard	4 Power BI pages
Focus	Performance · Products · Stores · Customers

Project scope

The case provides a set of analytical questions ranging from simple KPI requests to more advanced retention, store and basket analyses. These questions are treated as ad-hoc prompts rather than a strict checklist. The dashboard is therefore organized around the business problems and the needs of different stakeholders.

1. Business context

Company

Xóm Retail is a global multi-channel retailer selling premium consumer products across North America, Europe, Asia and Australia. The business operates hundreds of physical stores in 20+ countries together with an online channel. Its catalog contains 2,500+ SKUs across eight main categories: Computers, Cell Phones, Home Appliances, Audio, TV and Video, Games and Toys, Cameras and Camcorders, and Music.

Operating model

- Merchandising HQ: based in the US and responsible for catalog, pricing and vendor negotiations.
- Regional operations: responsible for stores, staffing and logistics.
- Customer data: centralized customer master containing demographic and geographic information.

Business pain points

- Delivery delay: Operations wants to understand which countries have slower online fulfillment.
- Customer churn: CRM does not have a cohort-based view of whether customers return after their first purchase.
- Store cannibalization: the business suspects that opening new stores may reduce revenue at existing stores.

2. Stakeholders and expected use

Stakeholder	What they need from the analysis
Head of Merchandising	Product performance, margin, assortment and cross-selling opportunities.
VP Store Operations	Delivery performance, store productivity and expansion risks.
Head of CRM	Retention, repeat purchase, customer value and VIP customers.
Head of Finance	Revenue, gross profit and margin performance.
COO	A concise view of overall performance and the most important business issues.

Deliverables

- Prepare and validate the retail data for analysis.
- Answer business questions using SQL-based analytical workflows.
- Build an interactive Power BI dashboard with consistent filters and navigation.
- Summarize the findings and translate them into practical business actions.

3. Dataset & Data modeling

Table	Rows	Contents
customers	~15.3K	Customer demographics and geography
products	~2.5K	Products, brands, category, cost and price
sales	~62.9K	Order lines, dates, customer/store/product keys and quantity
stores	67	Country, store area and opening date

SALES

Retail sales transactions across multiple countries.

Column	Data Type	Allow NULL
order_number	nvarchar(50)	NO
line_item	int	NO
order_date	date	YES
delivery_date	date	YES
customer_key	int	YES
store_key	int	YES
product_key	int	YES
quantity	int	YES

PRODUCTS

Product catalog containing category hierarchy, cost, and pricing details.

Column	Data Type	Allow NULL
product_key	int	NO
product_name	nvarchar(255)	NO
brand	nvarchar(100)	YES
color	nvarchar(50)	YES
unit_cost_usd	decimal	YES
unit_price_usd	decimal	YES
subcategory_key	int	YES
subcategory	nvarchar(100)	YES
category_key	int	YES
category	nvarchar(100)	YES

CUSTOMERS

Customer demographics and geographical information.

Column	Data Type	Allow NULL
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customer_key	int	NO
gender	nvarchar(100)	YES
name	nvarchar(100)	NO
city	nvarchar(50)	YES
state_code	nvarchar(100)	YES
state	nvarchar(50)	YES
zip_code	nvarchar(20)	YES
country	nvarchar(50)	YES
continent	nvarchar(50)	YES
birthday	date	YES

STORES

Physical store locations, operational area, and opening dates.

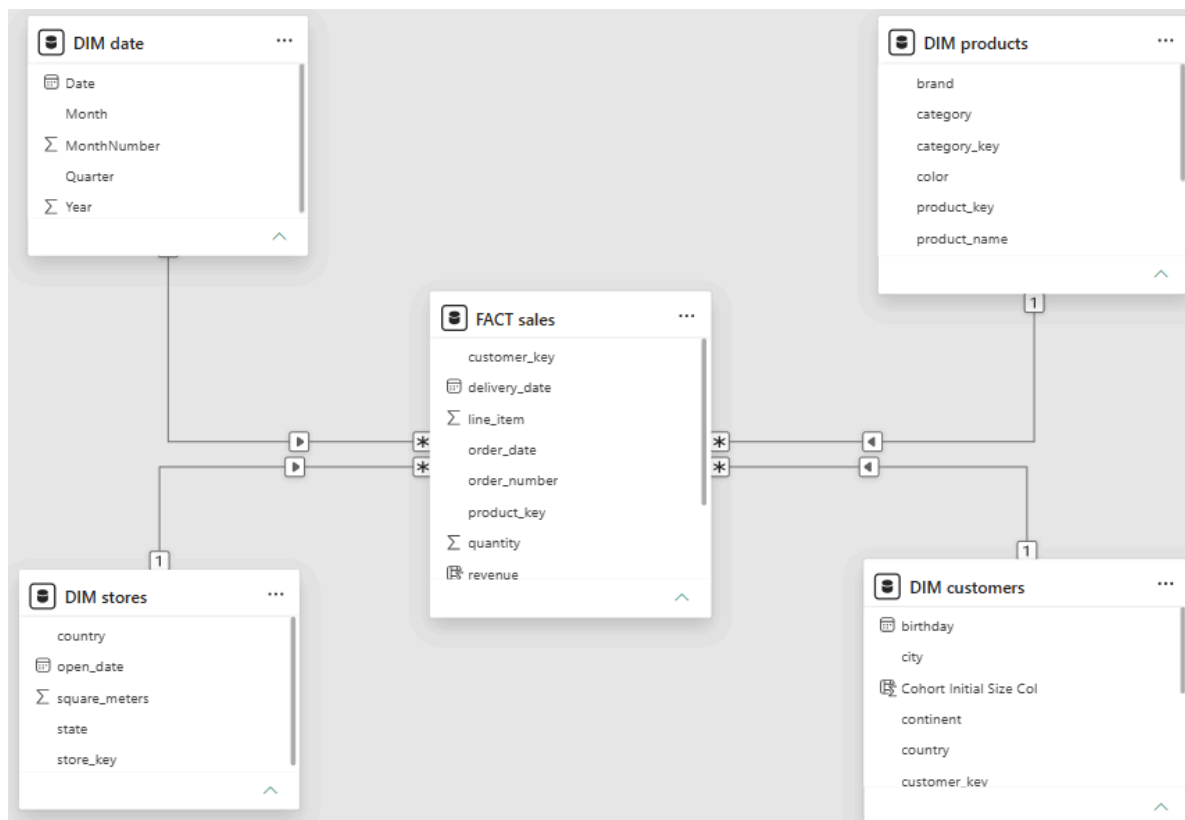
Column	Data Type	Allow NULL
store_key	int	NO
country	nvarchar(50)	YES
state	nvarchar(50)	YES

square_meters	decimal	YES
open_date	date	YES

Key relationships

The data model is designed using a **Star Schema** optimized for Power BI:

- **FACT sales** is at the center, containing transaction metrics.
- **DIM date**: Linked to **FACT sales** via order_date (1-to-Many).
- **DIM products**: Linked to **FACT sales** via product_key (1-to-Many).
- **DIM stores**: Linked to **FACT sales** via store_key (1-to-Many).
- **DIM customers**: Linked to **FACT sales** via customer_key (1-to-Many).



Important business rule

Average delivery time is calculated only for records with delivery dates, corresponding to eligible online/fulfillment transactions. The broader transaction data should not be interpreted as a complete channel-level view of customer behavior.

4.1 Overview

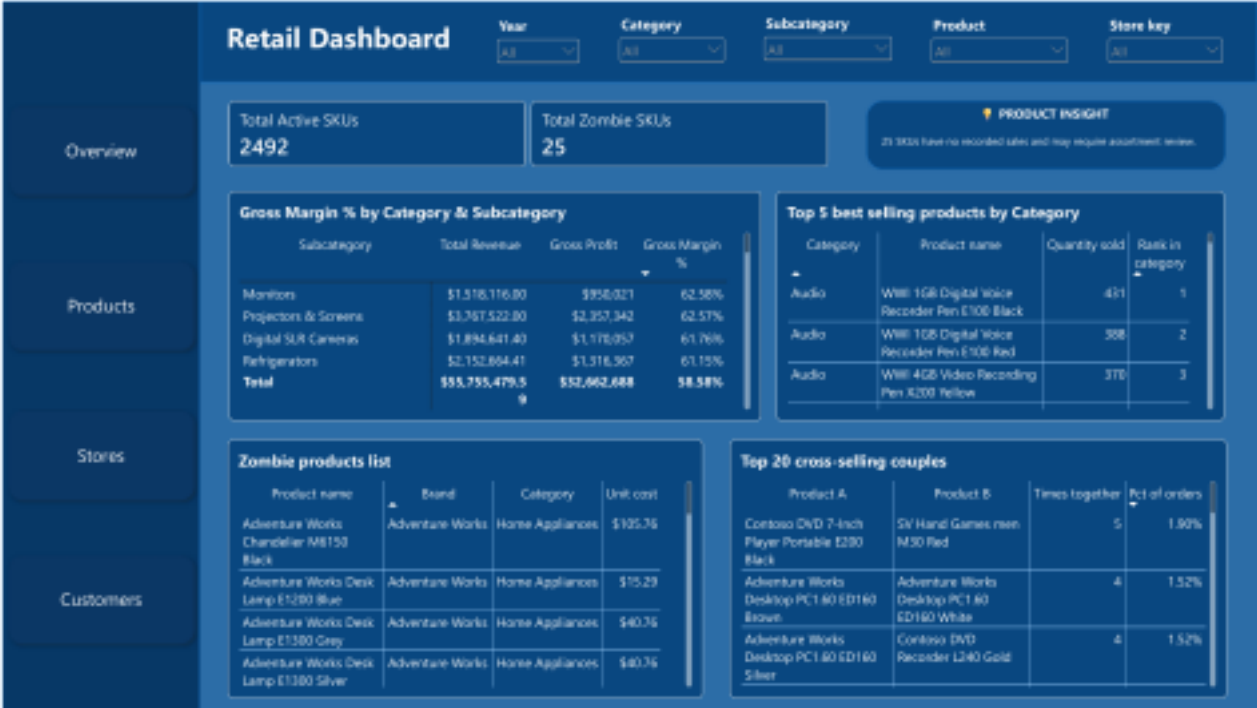
Executive view of overall performance, geography and top products.



The page starts with five KPIs, followed by revenue trend, store footprint, active customers by country, and top-selling products.

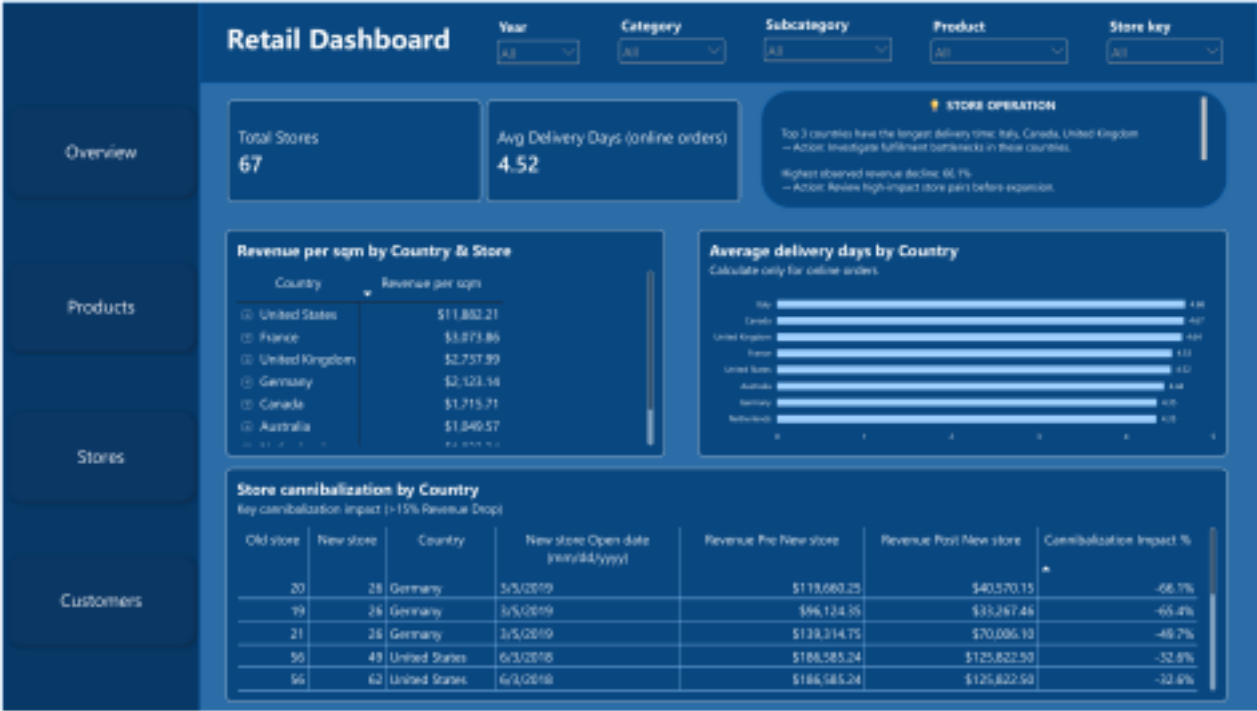
4.2 Products

Merchandising view of assortment, margin and product relationships.



4.3 Stores

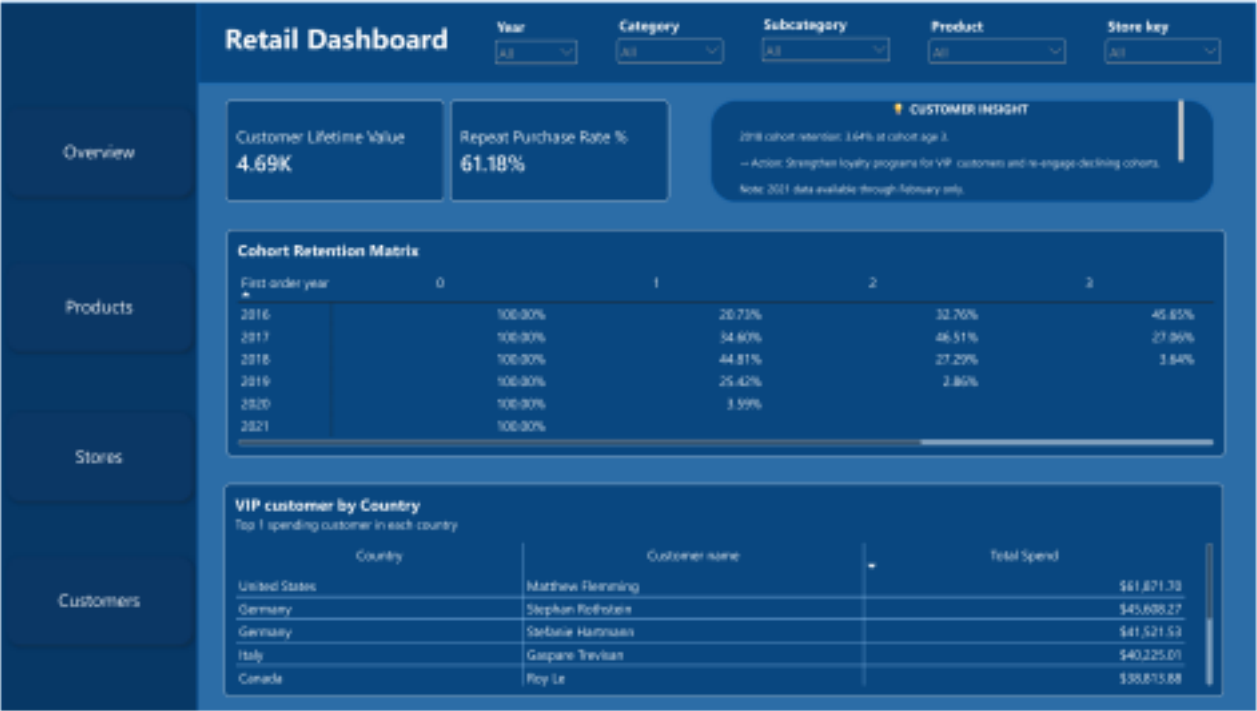
Operations view of store productivity, delivery and expansion risk.



Revenue per square meter provides a productivity view, while delivery time and store-pair analysis surface operational exceptions.

4.4 Customers

CRM view of customer value and retention.



CLV and repeat purchase rate are paired with cohort retention and top-spending customers to support customer strategy.

5. Key findings

Area	Finding	Business implication
Overall performance	Revenue is \$55.76M with a gross margin of 58.58% and AOV of \$2.12K.	This provides the baseline for the executive overview.
Product assortment	25 SKUs are classified as zombie products because they have no recorded sales.	These products should be reviewed alongside inventory and availability data before assortment decisions are made.
Delivery	Italy has the highest average delivery time among the countries shown at 4.68 days, followed by Canada at 4.67 days and the UK at 4.64 days.	Operations can use these markets as starting points for investigating fulfillment and carrier performance.
Store cannibalization	The largest observed impact in the dashboard is a 66.1% revenue decline for an affected German store pair.	High-impact store pairs should be reviewed before making further expansion decisions.
Customer retention	The 2018 cohort has 3.64% observed retention at cohort age 3.	Loyalty programs can be strengthened for VIP/high-value customers, while declining purchase activity can be targeted with re-engagement.
Geographic concentration	The US has the largest store footprint with 24 stores and the largest active customer base at 5.2K.	The US can be used as a benchmark when comparing other markets.

6. Recommendations

Merchandising

- Review zombie SKUs and combine sales status with inventory information before rationalizing the assortment.
- Use gross-margin and best-seller views to identify products that contribute strongly to revenue and profit.

Store operations

- Investigate the slowest delivery markets using carrier, fulfillment and country-level operational data.
- Flag large store revenue declines after new-store openings for deeper pre/post analysis.

CRM

- Use loyalty programs to retain VIP/high-value customers.
- Build targeted re-engagement journeys for customers whose purchase activity is declining.

7. Data limitations and next steps

Limitation	What it means
2021 coverage	The dataset contains 2021 data only through February. Later cohort-age periods are therefore incomplete.
Delivery analysis	Delivery time can only be calculated where delivery_date is available.
Cannibalization	The pre/post revenue comparison is observational. It does not by itself establish that a new store caused the decline.
VIP definition	The dashboard uses top-spending customers as the VIP view. A production CRM process should define VIP status explicitly.
CLV	The exact CLV formula and time horizon should be documented before the metric is used for operational decisions.
Data validation	Store-pair calculations and ranking logic should be checked for duplicate joins before the results are used for expansion decisions.

Next analytical steps

- Add inventory/on-hand data to quantify the financial exposure of zombie SKUs.
- Use controlled pre/post windows and additional market controls for a stronger cannibalization study.
- Build RFM/value tiers and measure loyalty-program impact through experiments.
- Extend channel-level analysis to compare online and physical customer behavior.